Muhammad Tahir Zia

PASSCO Society, Lahore, Pakistan | itstahir256@gmail.com | +92-304-5744237
Linkedin.com/in/Muhammad-Tahir-Zia | GitHub.com/TahirZia-1 | Portfolio

Summary

Aspiring Digital IC & RTL Design Engineer with hands-on experience in SystemVerilog and advanced RISC-V microarchitecture. Proficient in RTL design and simulation. Passionate about FPGA prototyping and high-performance digital logic design.

Education & Training

Digital IC Design & Verification (INSPIRE) Bootcamp, Pakistan Software Export Board (PSEB) at GIKI — May 2026 – Present

- **Focus:** Intensive specialized training in RTL design, advanced processor microarchitecture, and functional verification methodologies.

BS in Computer Engineering, Ghulam Ishaq Khan Institute of Engineering Sciences and Technology, Topi, Pakistan — Sept 2021 – June 2025

- **Coursework:** Fundamentals of Logic Design, Computer Organization and Assembly Language, Microprocessor Interfacing, Computer Communication and Networking, Signals & Systems, Digital Systems Design.

Technical Skills

Languages: Verilog, SystemVerilog, C/C++, RISC-V.

Tools: Vivado, Icarus Verilog, EDA Playground, Nexys A7 FPGA, Arduino UNO, MATLAB.

Projects

32-bit RISC-V Pipelined Processor Core – *SystemVerilog, Vivado* (GitHub Repository)

- Architected an RV32I 5-stage pipelined processor, evolving from a foundational single-cycle datapath.
- Engineered a custom Hazard Unit to dynamically resolve data and control hazards via EX/MEM & MEM/WB forwarding, load-use stalling, and branch flushing.
- Working on a branch predictor by utilizing 8-bit Global History Register and 256-entry Pattern History Table.

FPGA Morse Code Encoder – *Verilog, Vivado, Nexys A7* (GitHub Repository)

- Converts digits into its Morse code using an FSM based approach, displayed using LED lights. 1 second for dot and 3 seconds for Dash. Can be toggled into different modes for storing numbers in FIFO register as well.

Digital Clock – *Verilog, Vivado, Nexys A7* (GitHub Repository)

- Verilog implementation of a 24-hour digital clock. The design displays hours, minutes, and seconds on a 7-segment display that can be easily integrated into various FPGA development boards like Nexys A7.

FPGA Lab Design & Verification Work (GitHub Repository)

- Designed a **4-bit array multiplier**, **6-bit csa multiplier**, in verilog.
- Designed Parallel Prefix Adders like **Kogge Stone adder**, **Sklansky adder**, **Brent Kung adder**, in verilog.
- Designed an **FSM and corresponding code** to activate the signal generator activated in two different modes. **Short Blink Mode & Long Blink Mode** in Verilog using Vivado on Nexys A7.

Final Year Project

BlockConnect (Social Media Application using Blockchain) — May 2024 – Jun 2025

- Developed a decentralized social networking layer by using Ethereum blockchain focused on data sovereignty and user privacy. Leveraged the architecture to eliminate the third-party data commoditization.